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For Science and Technology

BMC

Respiratory Failure/Acute Respiratory Distress Syndrome

An approach to a Case of Cyanosis

Dr. Essam Kotb
Associate Professor
Internal Medicine Department

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Learning Objectives

Knowledge:

- Distinguish between Type 1 and Type 2 respiratory failure, as well as acute versus chronic presentations.
- Explain the underlying mechanisms of Type 1 and Type 2 respiratory failure, as well as the diffuse alveolar damage (DAD) in ARDS.
- Recognize the direct and indirect causes of ARDS and respiratory failure.
- Describe the diagnostic evaluation for respiratory failure and the Global Definition / Berlin criteria for ARDS.
- Contrast central versus peripheral cyanosis based on clinical presentation, underlying mechanisms, and differential diagnoses.

Skills:

- Analyze ABG results to determine the type and severity of respiratory failure and evaluate oxygenation ratios.
- Formulate a systematic diagnostic workup using bedside tools, laboratory tests, chest X-rays, CTs, and point-of-care ultrasound (POCUS) to identify underlying etiology.
- Select appropriate oxygen delivery modalities and apply evidence-based management algorithms (ABCDE approach).
- Apply principles of lung-protective ventilation in ARDS management, including lower tidal volumes, low plateau pressures, and adjusting positive end-expiratory pressure (PEEP).

Values:

- Recognize life-threatening signs of acute respiratory distress early to initiate prompt resuscitation and emergency airway management.
- Appreciate the clinical importance of standardized trial guidelines (such as ARDSNet protocols) to reduce ventilator-induced lung injury and improve survival rates.
- Maintain institutional rules, regulations, and ethical clinical standards during emergency and critical care training.

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- Type 1 respiratory failure (definition, Types, pathogenesis, clinical presentation, diagnosis, management)
- Type 2 respiratory failure (definition, pathogenesis, clinical presentation, diagnosis, management)
- ARDS (definition, Types, pathogenesis, clinical presentation, diagnosis, management)
- Differential diagnosis and types of cyanosis (definition, Types, pathogenesis, clinical presentation, diagnosis, management)

Definition – Respiratory Failure

- **Respiratory failure:** it is the inability of the respiratory system to maintain adequate gas exchange, (normal Oxygen & Carbon dioxide level), it is either Acute or Chronic
  - Hypoxemic respiratory failure: defined as a $P_aO_2 < 60$ mm Hg
  - Hypercapnic respiratory failure: defined as a $P_aCO_2 > 45$ mm Hg (Normal 35 – 45)

- **Respiratory arrest**: the complete cessation of breathing in patients with intact pulse

Classification by duration

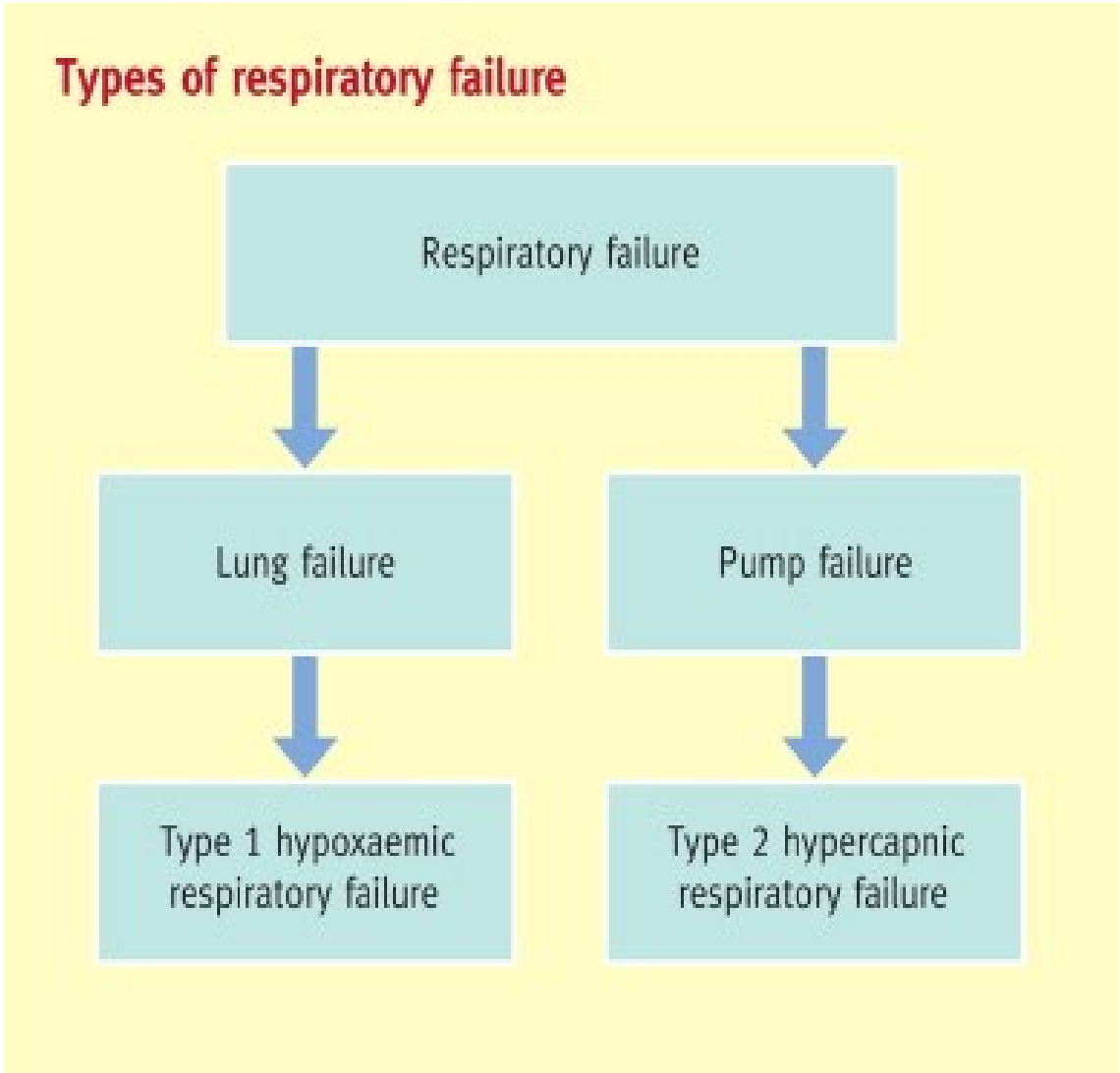
- **Acute respiratory failure**: develops over minutes to hours as a result of an acute illness or insult
- **Chronic respiratory failure:** longstanding respiratory failure resulting from chronic illness (COPD, ILD, obesity hypoventilation syndrome)

Classification

Types of respiratory failure by gas exchange abnormality:

- **Type 1 (hypoxemic respiratory failure)**
  - Definition: Respiratory failure characterized by **hypoxia and normocapnia or hypocapnia**
  - **PaO_2**: ↓ (< 60 mm Hg)
  - **$PaCO_2$**: Normal or ↓ (< 33 mm Hg)

- **Type 2 (hypercapnic respiratory failure)**
  - Definition: Respiratory failure characterized by **hypercapnia and normocemia or hypoxemia**
  - **PaO_2**: Normal or ↓ (< 80 mm Hg)
  - **$PaCO_2$**: ↑ (> 50 mm Hg)



Pathophysiology – Respiratory Failure

Type I:

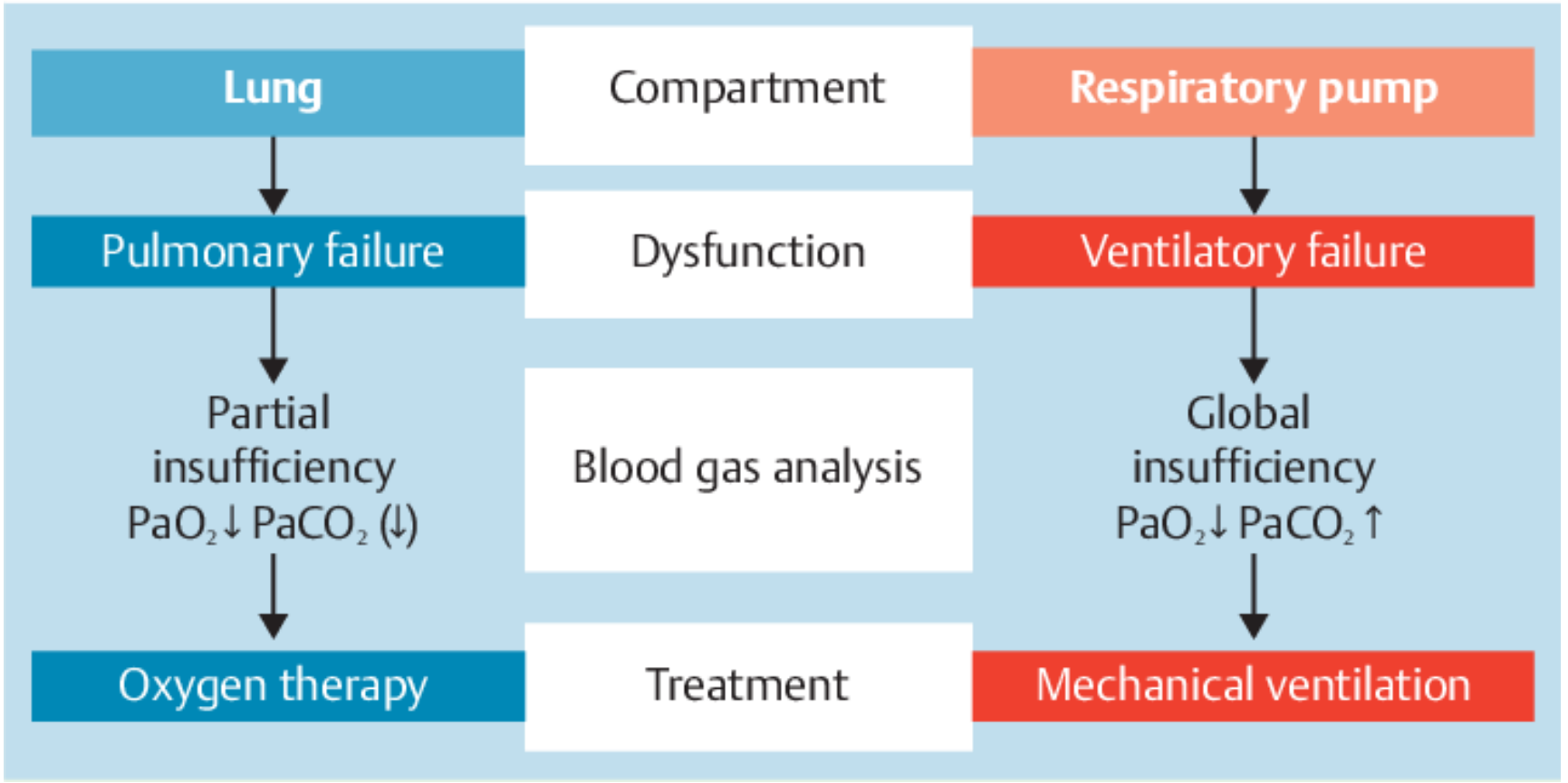
- Pulmonary disease (asthma, pneumonia)→ Impaired ventilation, perfusion of that under ventilated area → hypoxia & CO2 enter blood → increased ventilation of neighboring regions of normal lung →↑Co2 execration, correcting Co2 to normal.
- It cannot correct O2 because Hb passing though is fully saturated

Type II:

- Pulmonary disease (Acute sever asthma, AECOPD) → severe ventilation perfusion miss match→ insufficient lung to correct CO2 → ↑Co2 level in blood

Classification – Respiratory Failure

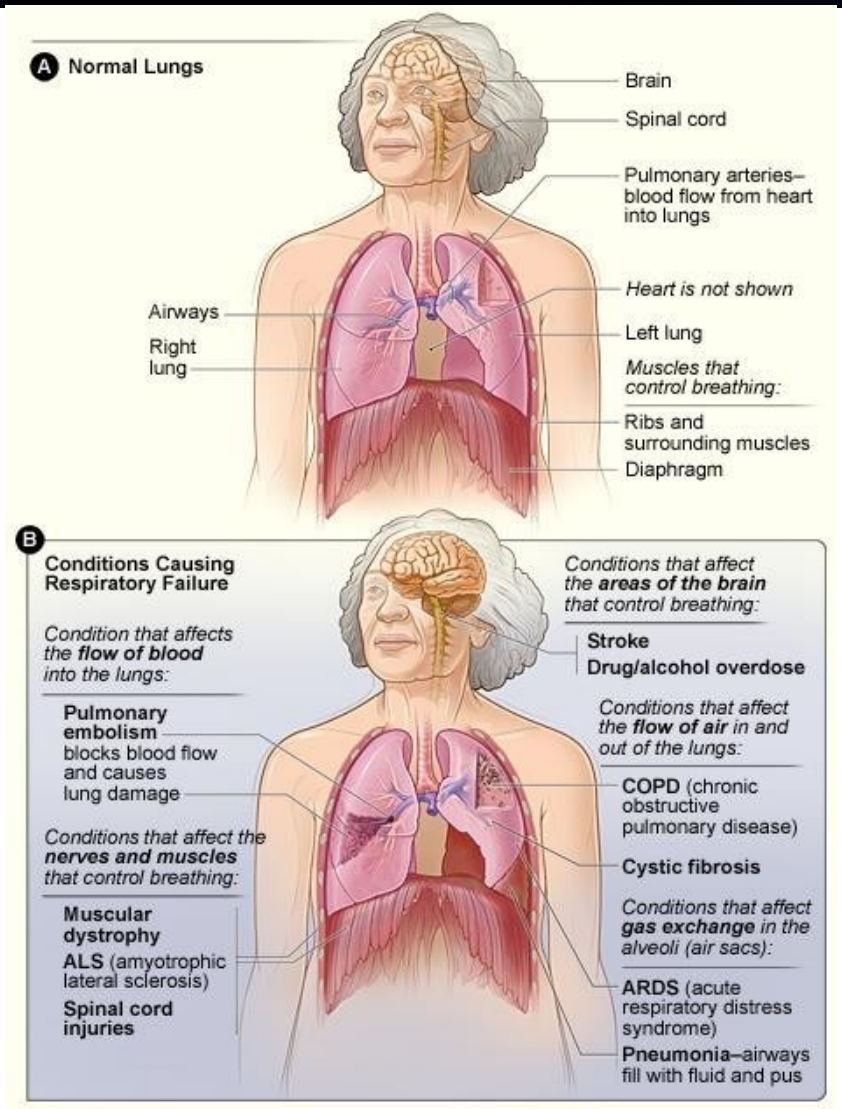
Type I Vs. Type II



Etiology – Respiratory Failure

Causes of hypoxemia: (TYPE I)

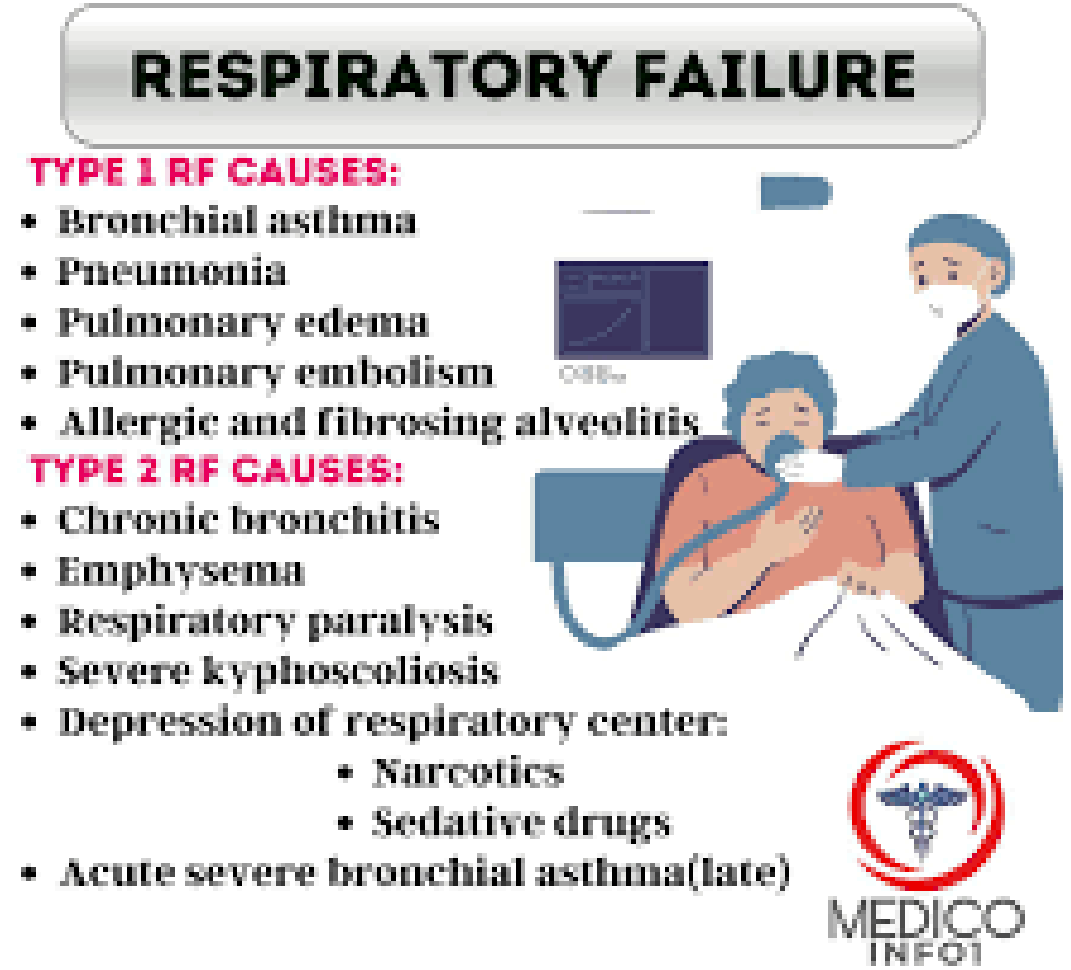
- impaired alveolar diffusion (edema, Pneumonia, IPF)
- Right-to-left shunt
  - Pulmonary right-to-left shunt (ARDS, tension pneumothorax)
  - Intracardiac right-to-left shunt (Eisenmenger syndrome)
- V/Q mismatch (pneumonia, embolism)
- Decreased FiO_2 (Asphyxia)
- Hypoventilation



Etiology – Respiratory Failure

Causes of hypercapnia: (TYPE II)

- Airway obstruction and/or increased physiologic dead space (AECOPD, Asthma, drowning, OSA)
- CNS depression (Opioid intoxication, benzodiazipines)
- Decreased chest wall compliance (Rib fracture, sleep apnea, ankylosing spondilitis)
- Respiratory muscle weakness (myasthenia gravis, Guillain-Barré syndrome, myopathies, ALS, high cervical spinal cord injury, poliomyelitis)
- Electrolyte disturbances



Clinical Features – Respiratory Failure

Symptoms:

Clinical features of hypoxemia

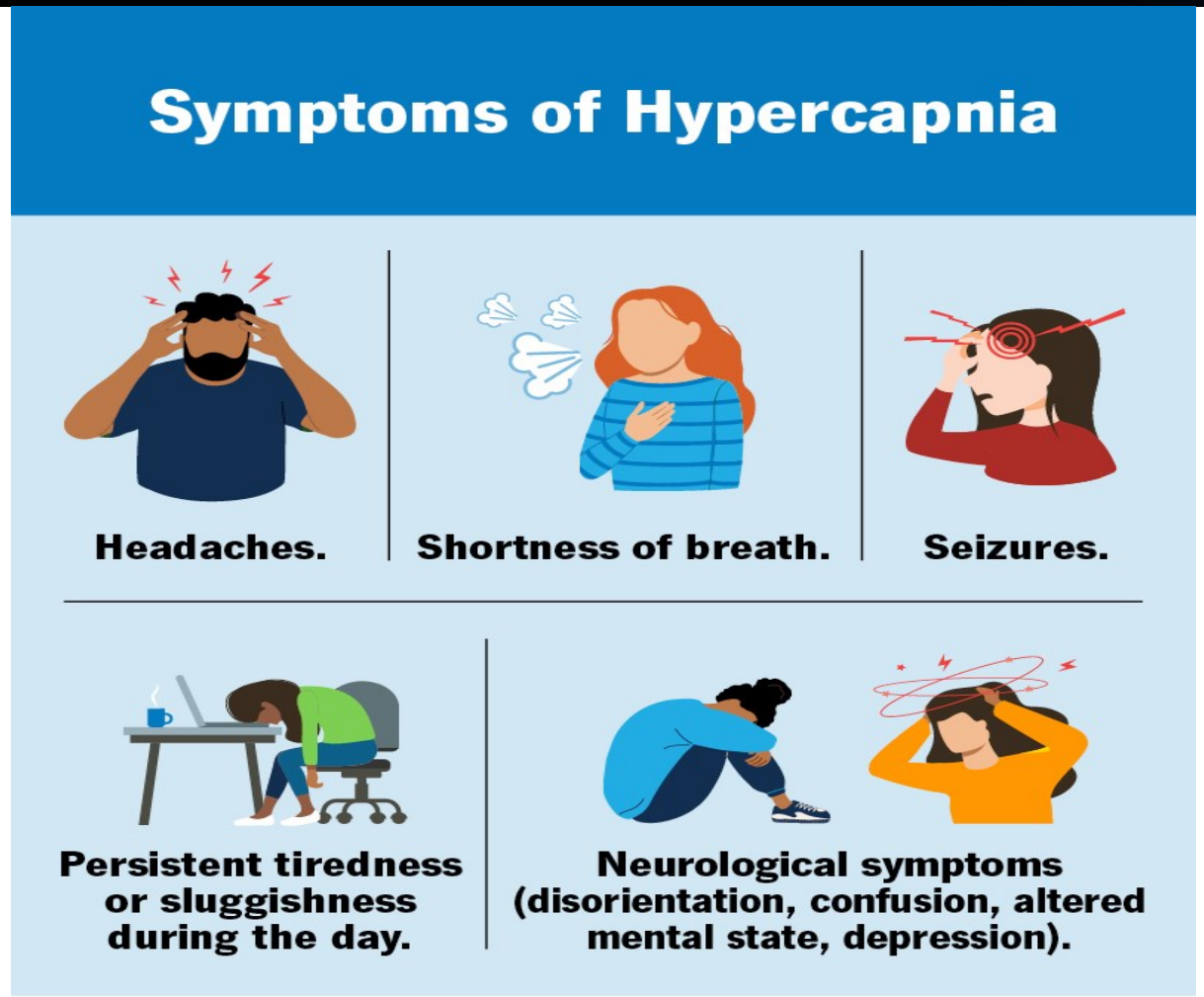
- Tachypnea, dyspnea
- Cyanosis
- Pleuritic chest pain
- Tachycardia, arrhythmia
- Altered mental status: confusion, somnolence, agitation



Clinical Features – Respiratory Failure

Clinical features of hypercapnia

- Hypopnea
- Headache, daytime sleepiness
- Anxiety
- Warm extremities
- Papilledema
- Asterixis
- Coma
- Paralytic ileus



Clinical Features – Respiratory Failure

Signs: clinical features of respiratory distress

- Inability to speak in complete sentences
- Increased work of breathing (increased respiratory effort)
  - Nasal flaring
  - Grunting
  - Pursed lips
- Use of accessory muscles of respiration
- Tachypnea & Hyperventilation
- Tachycardia
- Cyanosis
- Impaired mental state (due to fatigue or hypercapnia, or if cause of respiratory failure is CNS depression)

Diagnosis & Management

Diagnosis:

is clinical diagnosis. ABG analysis, rapid laboratory studies, and bedside imaging can help identify the type of respiratory failure and guide treatment of the underlying cause

- ABG
- CXR, or CT chest
- CBC and metabolic panel
- If cardiogenic pulmonary edema is suspected, consider cardiac enzymes.
- Diagnosis of the cause: - CXR-POCUS-CT-CTA

Diagnosis & Management

Approach

- Use ABCDE approach
- Provide immediate respiratory support according to the cause and severity of respiratory failure.
- Identify and treat rapidly reversible causes of respiratory failure. (Tension Pneumothorax, pulmonary edema, anaphylaxis, status asthmaticus, aspiration of foreign body
- Focused clinical evaluation.
- Obtain initial diagnostics: e.g., ABG, routine laboratory studies, ECG, CXR, POCUS
- Consider advanced diagnostic testing based on initial findings.
- Treat the Cause.

Diagnosis & Management

Monitoring and disposition

- Admit to the ICU for close monitoring with respiratory and/or hemodynamic support.
- Monitor respiratory and hemodynamic parameters.
- Follow serial ABGs.
- Consider repeating laboratory studies and/or chest imaging if the patient's clinical status changes.

Respiratory support

- **Airway management**: Secure the airway.
- **Rescue breaths**: indicated immediately for patients in respiratory arrest
- **Oxygen therapy**
  - Indicated for SpO_2 < 95%
  - Consider High flow nasal canula for hypoxemic respiratory failure.
- **Mechanical ventilation**
  - Consider indications for NIPPV and indications for invasive mechanical ventilation

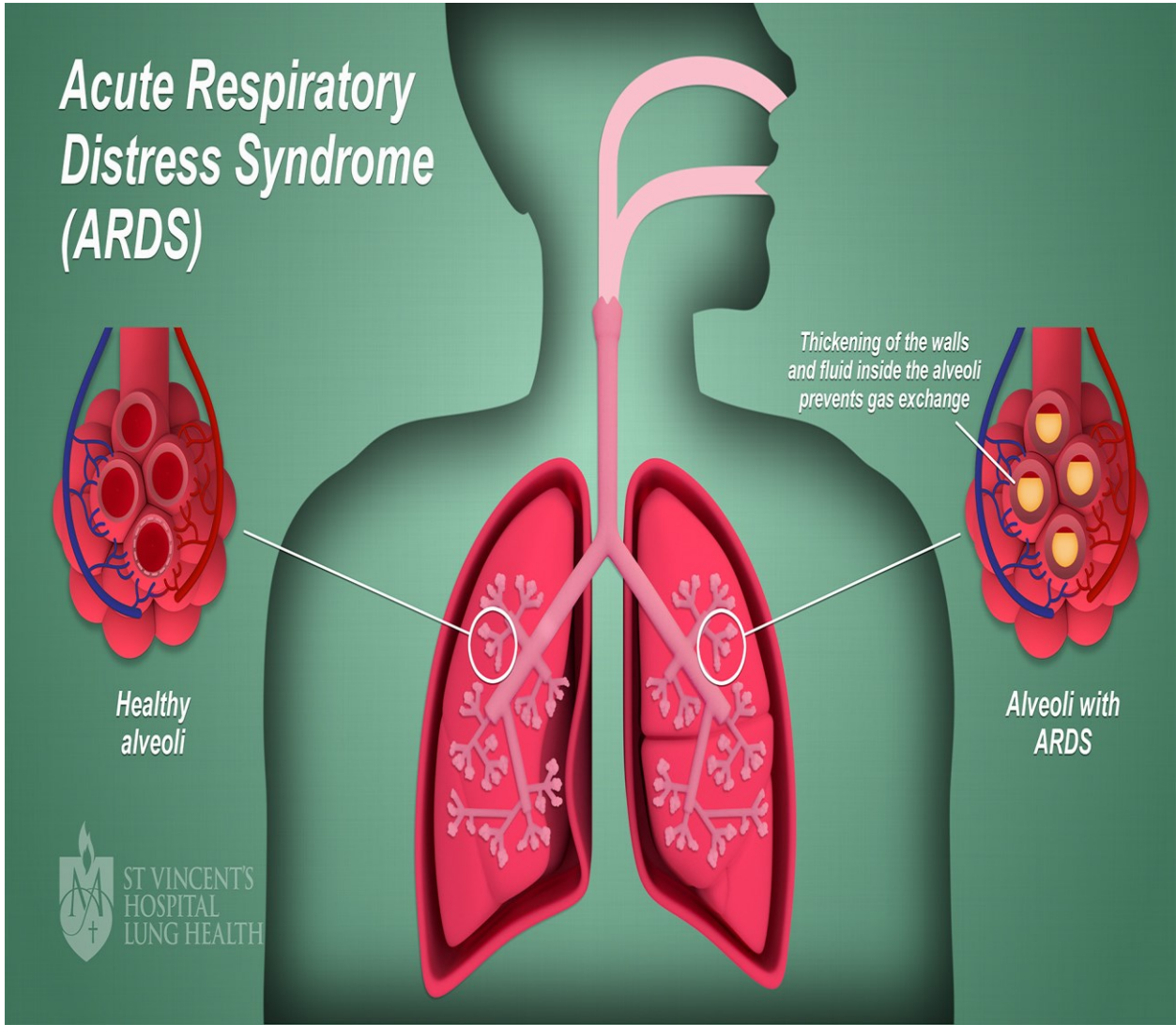


Acute Respiratory Distress Syndrome (ARDS)

Definition:

- A syndrome of acute respiratory failure characterized by hypoxemia and bilateral pulmonary infiltrates that cannot be fully explained for by heart failure or fluid overload.

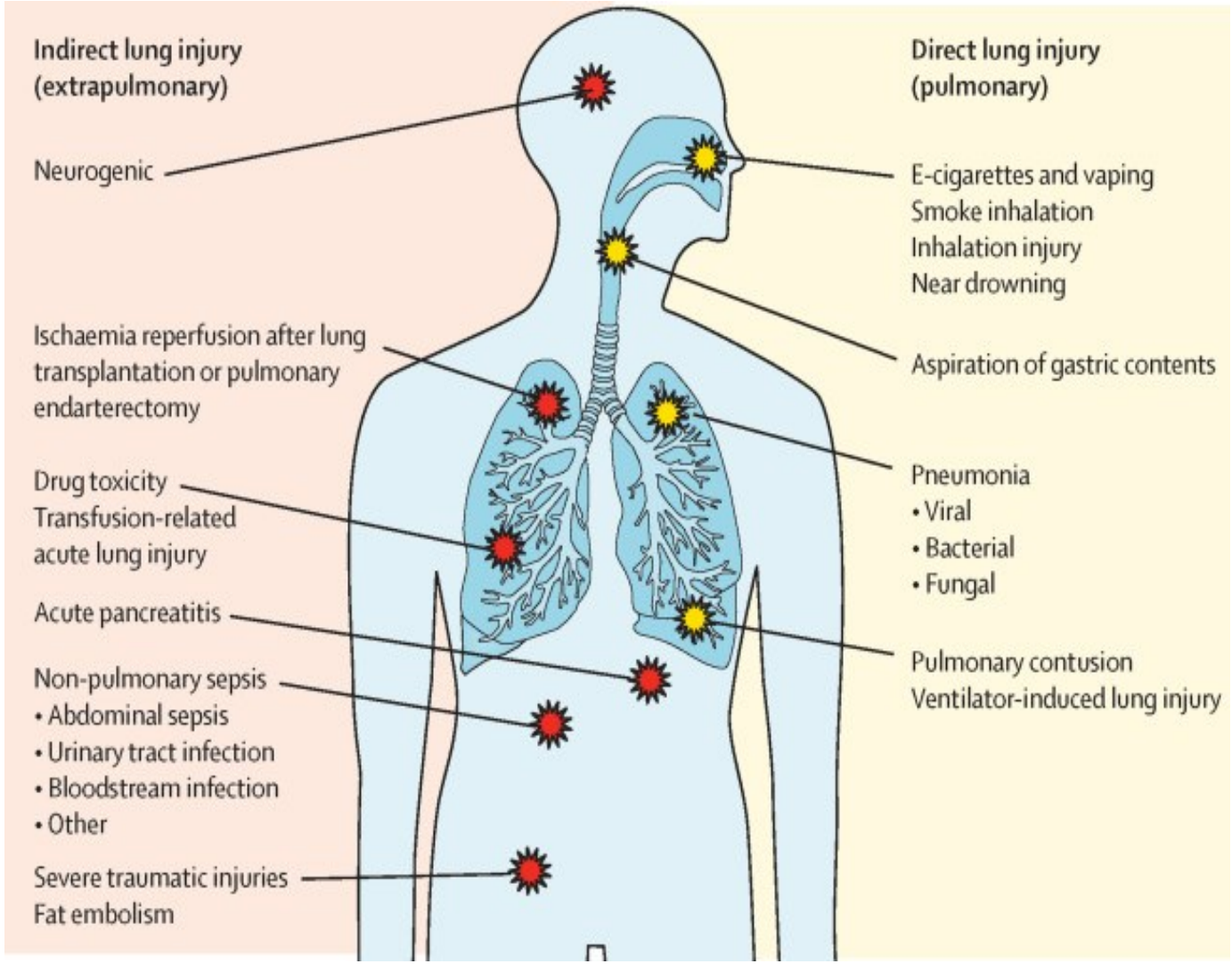
- Not a primary disease
- A disorder that arises due to other conditions → widespread inflammatory process.



Acute Respiratory Distress Syndrome (ARDS)

Systemic Causes:

- Sepsis (most common cause)
- Trauma
- Shock
- Massive blood transfusion (TRALI)
- Acute pancreatitis
- Hematopoietic stem cell transplantation
- Medications
- Recreational drug overdose (e.g., cocaine)

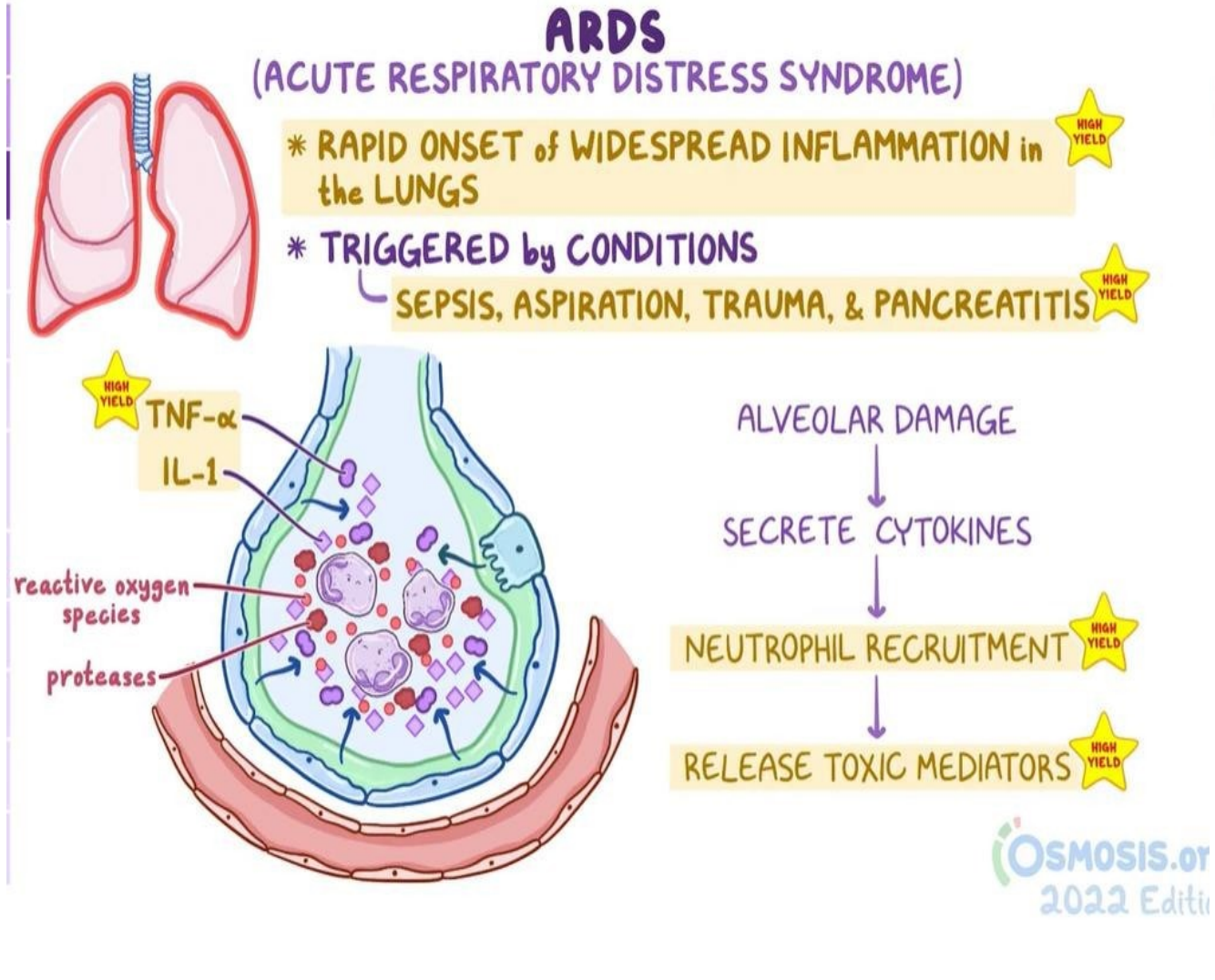


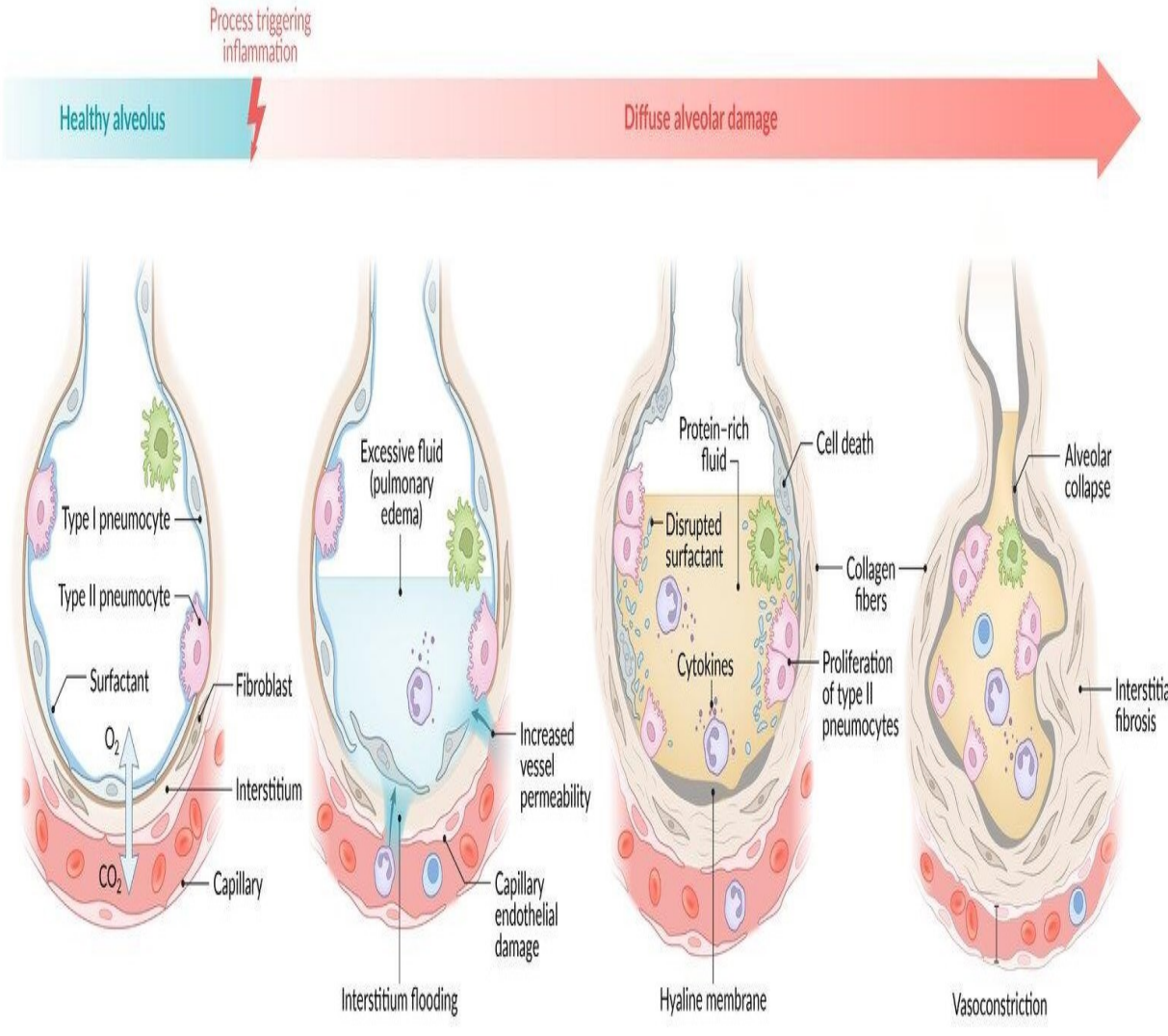
Acute Respiratory Distress Syndrome (ARDS)

Direct lung injury causes:

- Pneumonia
- Aspiration
- Inhaled toxins
- Pulmonary contusion
- Inhalation injury (e.g., inhalation of hyperbaric oxygen)
- Drowning incidents
- Fat embolism
- Amniotic fluid embolism
- Lung transplantation

Pathophysiology:





Acute Respiratory Distress Syndrome (ARDS)

Pathophysiology:

•Tissue damage (pulmonary or extrapulmonary) → release of inflammatory mediators → **sequestration of** neutrophils into alveoli → injury to alveolar capillaries and endothelial cells (**diffuse alveolar damage, DAD**) leading to:

- **Exudative phase**: excess fluid in interstitium and on alveolar surface → pulmonary edema with normal pulmonary capillary wedge pressure (**noncardiogenic pulmonary edema**) → **decreased lung compliance and respiratory distress**
- **Hyaline membrane formation**: exudation of neutrophils and protein-rich fluid into the alveolar space → formation of alveolar **hyaline membranes** → impaired gas exchange → hypoxemia →hyperventilation → respiratory alkalosis
  - Hypoxemia → chronic hypoxic pulmonary vasoconstriction → pulmonary hypertension
  - **Damage** to type I and type II pneumocytes → decrease in surfactant → alveolar collapse → intrapulmonary shunting
- **Organizing phase (late stage)**: **proliferation of type II pneumocytes** and infiltration of fibroblasts → **progressive interstitial fibrosis**

Acute Respiratory Distress Syndrome (ARDS)

Signs & Symptoms:

- Acute dyspnea
- Tachypnea and tachycardia
- Cyanosis
- Diffuse crackles
- Fever, cough, and chest pain may also be present.

Acute Respiratory Distress Syndrome (ARDS)

Liver Dysfunction

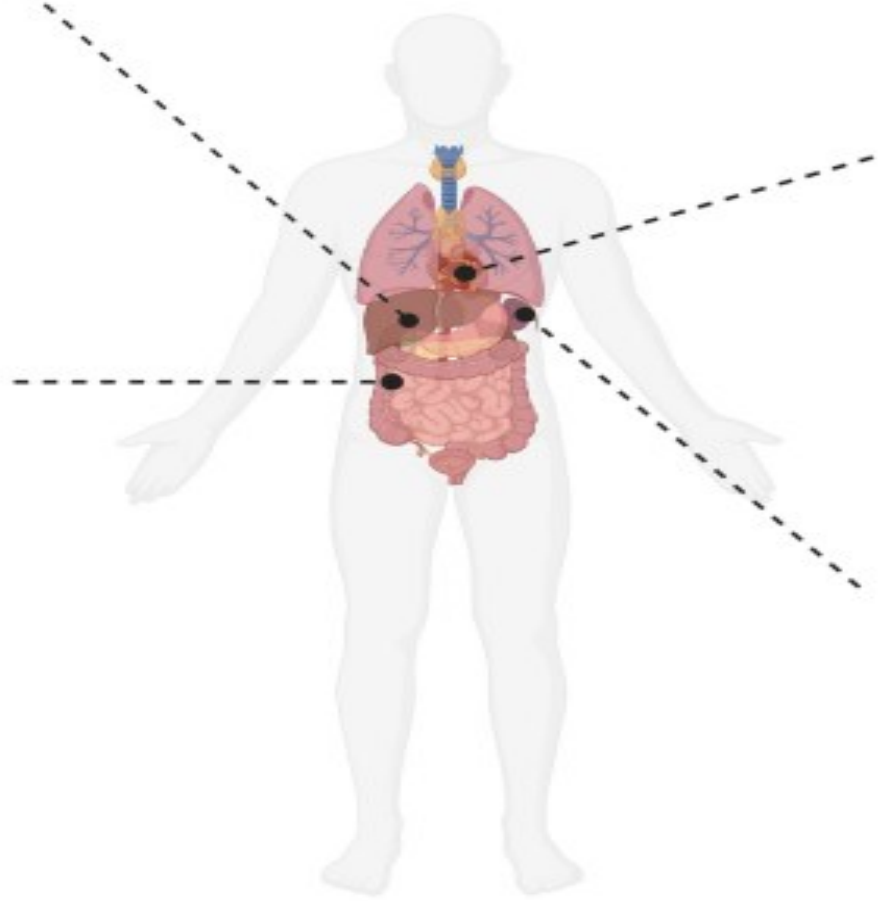
- Abnormalities of synthetic function
- Increased transaminases
- Hyperbilirubinemia
- Activation of coagulation cascade
- Production of acute-phase proteins
- Hypoalbuminaemia
- Hypoglycaemia

Gastrointestinal Dysfunction

- Acute stress ulcerations
- Bleedings
- Ileus

Haematologic Dysfunction

- Thrombocytopenia
- Consumptive coagulopathy
- Anaemia



Neurologic Dysfunction

- Alterations in level of conciousness
- Confusion
- Phychosis
- Peripheral polyneuropathy

Cardiovascular Dysfuntion

- Refractory hypotension
- Dysrhythmias
- Increased capillary permeability producing oedema
- Loss of peripheral vascular tone
- Decreased cardiac output
- Myocardial infarction

Renal Dysfunction

- Reduced renal perfusion
- Acute kidney injury
- Oliguria
- Electrolyte abnormalities
- Decreased erythropoietin levels
- Vitamin D malabsortion
- Abnormal renin production

Acute Respiratory Distress Syndrome (ARDS)

Berlin Criteria:

ARDS New Global Definition 2023

new definition criteria	Classification		
	Mild	**Moderate**	**Severe**
Time to instalation	Up to seven days - known risk fator(s)		
Pulmonary edema	Not explained by cardiogenic edema or intravascular volume overload		
Radiologic features	Bilateral infiltrates on chest X-ray or CT or lung ultrasound (by a trained professional) (not explained by nodules, pleural effusion or atelectasis)		
Hypoxemia PaO_2/FIO_2**	201-300 with NIV/CPAP PEEP ≥ 5* or HFNO > 30l/min	101 - 200 com PEEP ≥ 5	≤ 100 com PEEP ≥ 5
Hypoxemia SpO_2/FIO_2	≤ 315 with SpO_2 ≤ 97%		

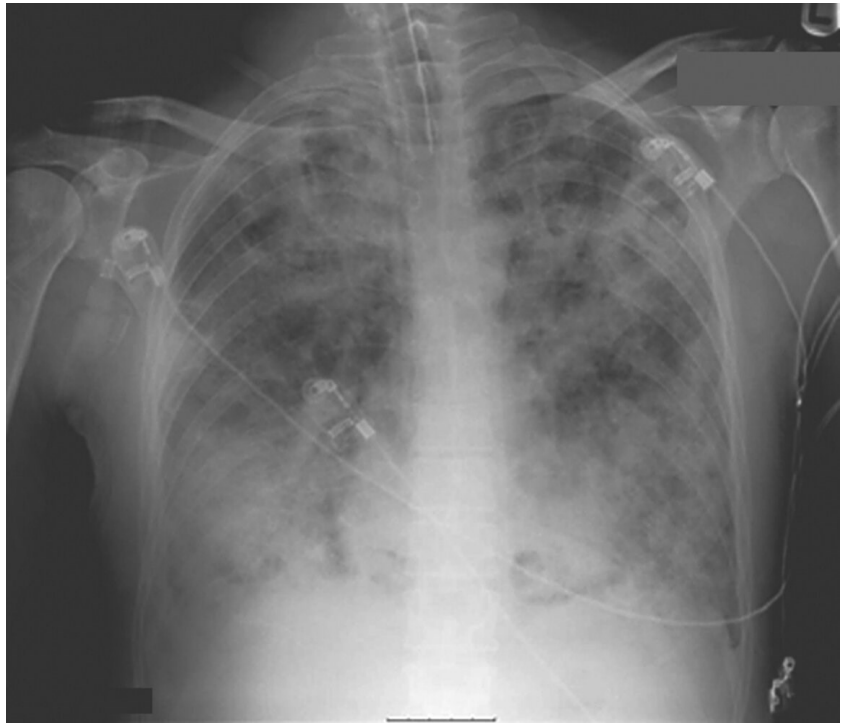
Am J Respir Crit Care Med 2023;207:A6229

Acute Respiratory Distress Syndrome (ARDS)

Diagnostics:

- CXR (diffuse bilateral pulmonary infiltrates)
- ABG (Hypoxemia $PaO_2 < 60$)
- Workup of underlying cause (e.g. septic workup, pancreatic enzymes, etc.)
- CT may be needed







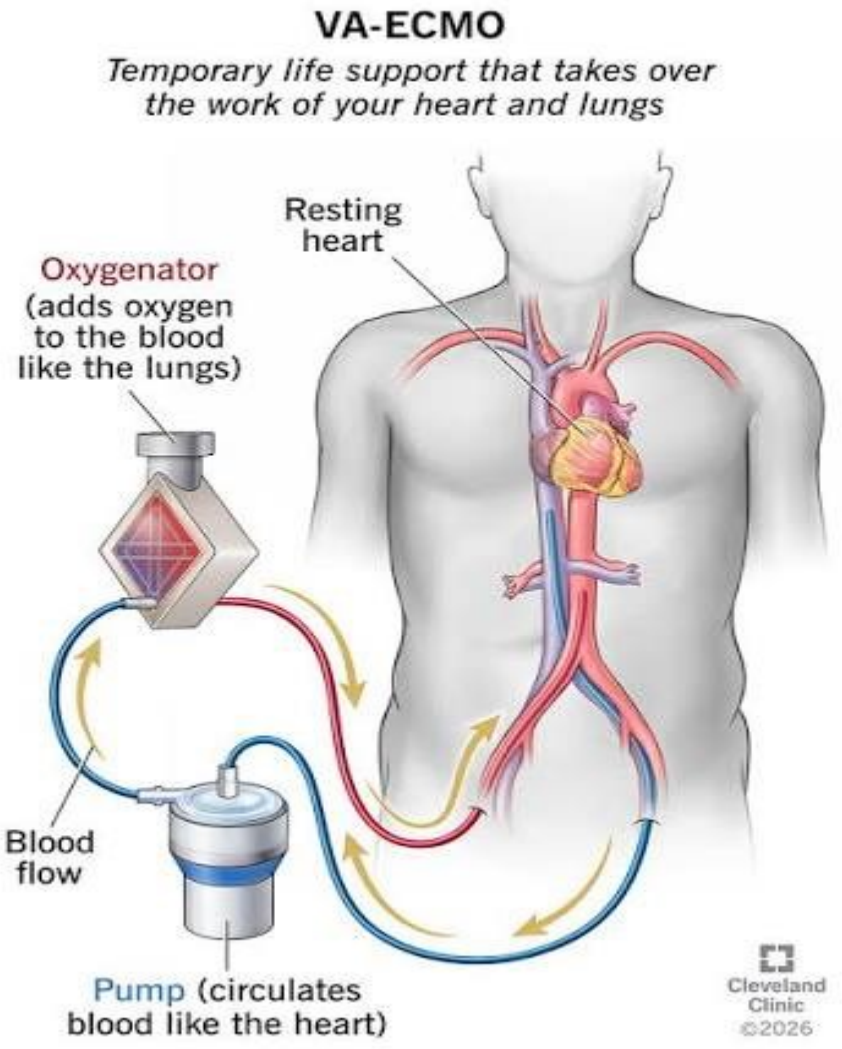
Acute Respiratory Distress Syndrome (ARDS)

Management:

- Early ICU admission & Treat the underlying cause
- Oxygenation
- Glucocorticoids in intubated patients
- Lung-protective ventilation per the ARDSNet study:
  - Low tidal volume
  - Low plateau pressure
  - High positive end expiratory pressure (PEEP)

Adjust therapy based on severity.

  - Moderate or severe: Use prone positioning and high PEEP strategy.
  - Early severe ARDS: Consider neuromuscular blockers.
  - Severe ARDS not responding to therapeutic adjustments: Consider ECMO.
- Fluid management (diuretics)



Approach to Cyanosis

Key History:

- Onset (age and acute vs chronic)
- Presence of chest pain
- Presence of dyspnea
- Smoking history
- History of lung disease, or heart failure
- Allergies
- Medications (beta blockers)

Key Physical Exam:

- Vital signs
- Pulse oximetry
- Heart
- Lungs
- Extremities (clubbing, cyanosis, edema)

Approach to Cyanosis

General Workup:

- Oxygen saturation
- ABG
- CBC
- ECG
- CXR
- Echo if Cardiac cause is suspected

Specific Workup:

- Sputum & blood culturs
- V/Q scan & CTA
- TTE
- Duplex Doppler

DDx:

CENTRAL CYANOSIS

DECREASED OXYGEN SATURATION

- Severe respiratory disease
- Pulmonary oedema
- Pulmonary embolism
- Congenital cyanotic heart disease

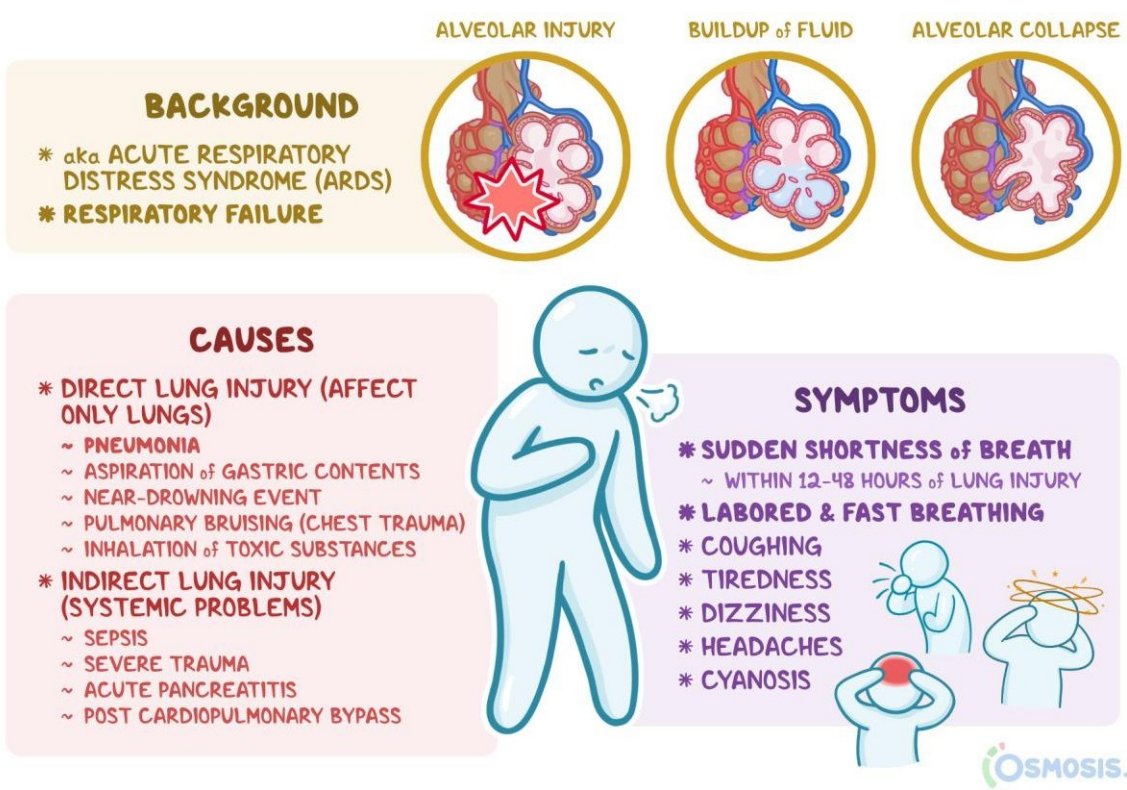
ABNORMAL HAEMOGLOBIN

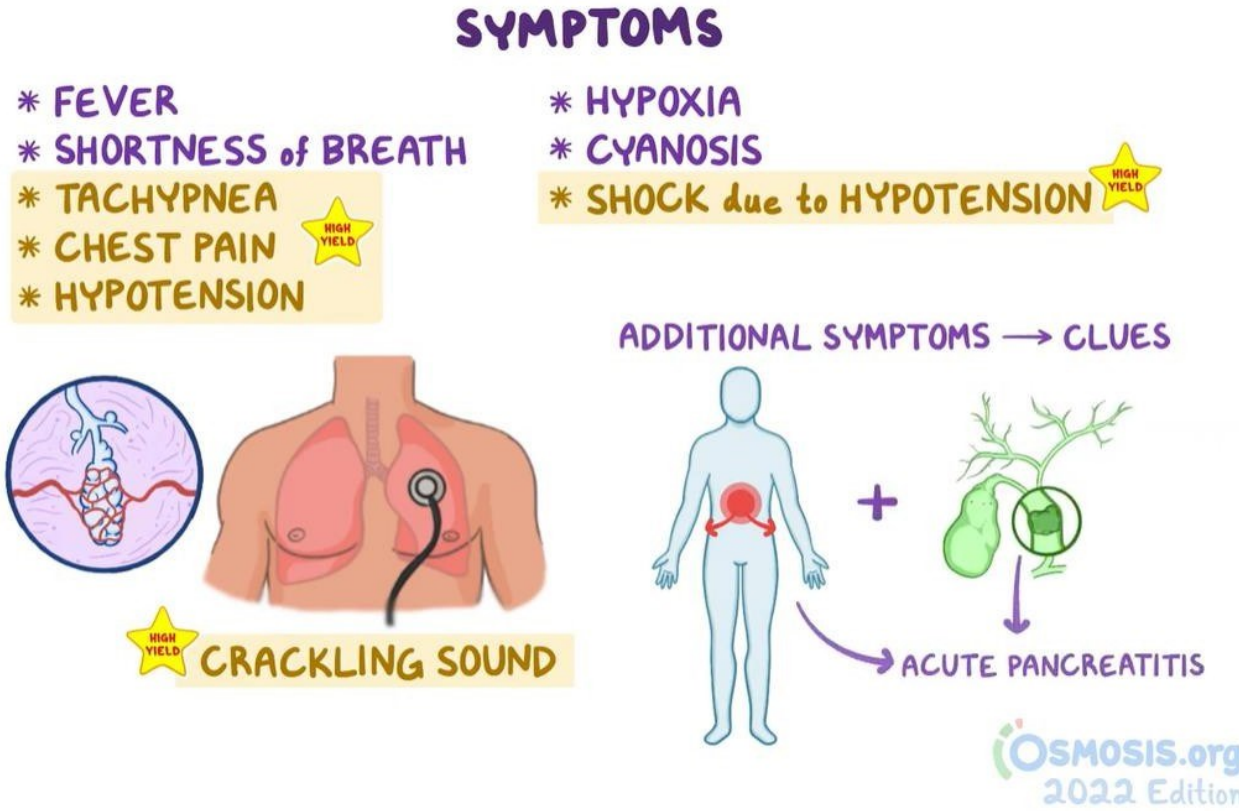
- Methaemoglobinaemia
- Sulphaemoglobinaemia

PERIPHERAL CYANOSIS

- All causes of central cyanosis
- Cold exposure
- Raynaud's phenomenon
- Arterial occlusion
- Venous occlusion
- Acrocyanosis

To Summarize





Case Scenario 1

A 63-year-old woman is brought to the emergency department because of a 2-day history of severe epigastric pain and nausea. She has a 20-year history of alcohol use disorder. Nine hours after admission, she becomes increasingly dyspneic and tachypneic. Pulse oximetry on supplemental oxygen shows an oxygen saturation of 81%. Physical examination shows diffuse lung crackles, marked epigastric tenderness, and a periumbilical hematoma. Laboratory studies show normal brain natriuretic peptide. An x-ray of the chest shows bilateral opacities in the lower lung fields. Which of the following pathomechanisms best explains this patient's pulmonary findings?

a. Alveolocapillary membrane leakage
b. Intrapulmonary left-to-right shunt
c. Increased production of surfactant
d. Embolic obstruction of pulmonary arteries
e. Inflammation of the bronchial mucosa
f. Increased pulmonary capillary pressure

Case Scenario 2

A 70-year-old man comes to the emergency room for worsening leg pain and a rash consistent with wet gangrene. He has no history of skin infections but has type 2 diabetes mellitus and has smoked 2 packs of cigarettes daily for 20 years. Three days after admission, he becomes increasingly hypoxic and tachypneic. He is emergently intubated and ventilatory support is initiated. He is 180 cm (5 ft 11 in) tall and weighs 90 kg (198 lb); BMI is 28 kg/m^2. His pulse is 112/min and blood pressure is 95/60 mmHg. The ventilator is set at a tidal volume of 540 mL, respiratory rate of 20/min, and positive end-expiratory pressure (PEEP) of 5 cm H_2O. On pulmonary examination, there are diffuse crackles. Cardiac examination shows no abnormalities. Hemoglobin concentration is 11.5 g/dL, serum lactate is 4.0 mmol/L, and serum creatinine is 2.5 mg/dL. An arterial blood gas checked 30 minutes after intubation shows a PCO_2 of 50 mm Hg and a PO_2 of 55 mm Hg on an F_IO_2 of 1.0. An x-ray of the chest shows new bilateral infiltrates. Which of the following is the most appropriate next step to improve tissue oxygen delivery in this patient?

a. Inhaled nitric oxide
b. Prone positioning of patient
c. Increase the tidal volume
d. Increase the positive end-expiratory pressure
e. Transfusion of red blood cells

Questions?



References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine 20th edition
- Amboss
- Cecil Essentials of Medicine 10th edition
- First Aid for the internal medicine board 4th edition
- Osmosis

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Thank You